Biology

Topic: Respiration in plants

Introduction and Do Plants Breathe?

1. Read the following statements and choose the incorrect option.
1) All living organisms need energy for carrying out daily life activities, be it absorption, transport, movement, reproduction or even breathing.
2) All the energy required for 'life' processes is obtained by oxidation of some macromolecules that we call 'food'.
3) Photosynthesis, takes place within the chloroplasts in both prokaryotes as well as eukaryotes.
4) The process of breathing is very much connected to the process of release of energy from food.

2. During the process of respiration,
1) Carbon dioxide and water are utilised, oxygen and energy are released as by-products
2) Carbon dioxide and water are utilised, water and oxygen are released as by-products
3) Oxygen is utilised, and carbon dioxide, water and energy are released as by-products
4) Oxygen and carbon dioxide are utilised, water and energy are released as by-products

3. Respiration is
1) Anabolic and exergonic
2) Catabolic and exergonic
3) Anabolic and endergonic
4) Catabolic and endergonic

4. Which of the following statement justifies the absence of respiratory organs in plants?
1) Each plant part takes care of its own gaseous exchange needs.
2) There is very little transport of gases from one plant part to another.
3) Roots, stem and leaves respire at rates far lower than animals do.
4) All of these.

5. Respiration differs from burning in which of the following?
1) Involvement of enzymes
2) Energy released in slow stepwise reactions
3) Oxidation of respiratory substrate
4) More than one option is correct

6. Energy can be obtained fastest from oxidation of _____
1) Glucose

2) Fat
3) Protein
4) Organic acids

7. Energy currency for cellular activities is
1) ATP
2) Protein
3) FAT
4) NAD

8. Read the following statements and choose the correct option.
Statement 1: The plant cell uses a strategy to catabolise glucose molecules in a way that not all the liberated energy goes out as heat.
Statement 2: The plant strategy is to oxidise glucose not in one step but in several small steps enabling some steps to be just large enough such that the energy released can be coupled to ATP synthesis.
1) Both statements are correct
2) Both statements are incorrect
3) Only statement 1 is correct
4) Only statement 2 is correct

9. In plants, gaseous exchange can occur through
1) Lenticels only
2) Stomata only
3) Both lenticel and stomata
4) Cuticle and stomata

Glycolysis

10. Which of the following cellular metabolic processes can occur in absence of oxygen?
1) Glycolysis
2) Kreb's cycle
3) TCA cycle
4) Electron transport

11. In glycolysis, conversion of glyceraldehyde-3 phosphate to 1,3-bisphosphoglyceric acid is catalyzed by which of the following enzyme?
1) Phosphofructokinase
2) Phosphoglycerate kinase
3) Triose phosphate dehydrogenase
4) Phosphoglyceromutase

12. In glycolysis, ATP is utilised during the conversion of
1) Glucose into glucose-6-phosphate and glucose-6- phosphate to fructose-6-phosphate

2) Glucose into glucose-6-phosphate and fructose-6- phosphate to fructose 1, 6-bisphosphate
3) Only during conversion of glucose into glucose-6- phosphate
4) Fructose-6-phosphate to fructose 1, 6-bisphosphate and phosphoenol pyruvate to pyruvic acid

13. All are true about glycolysis, except
1) It is a ten-step process.
2) Glucose and fructose are phosphorylated to give rise to glucose-6-phosphate by the activity of hexokinase.
3) The scheme of glycolysis was given by Gustav Embden, Otto Meyerhof and J. Parnas.
4) It produces one pyruvic acid molecule from each glucose molecule.

14. Identify the reaction of glycolysis which involves substrate level phosphorylation:
1) Glucos-6-phosphate-Fructose-6-phosphate
2) 1,3-bisphosphoglyceric acid-3-Phosphoglyceric acid
3) Triose-phosphate-—Triose bisphosphate
4) 2-Phosphoglycerate-Phosphoenolpyruvate

15. An irreversible step in glycolysis is/are catalyzed by
1) Hexokinase
2) Phosphofructokinase
3) Pyruvate kinase
4) All of these

16. First half of glycolysis is characterized as glucose activation phase or preparatory phase. It is marked by the
1) Synthesis of 2 ATPs by substrate level phosphorylation.
2) Phosphorylation of glucose molecule in the presence of enzyme hexokinase and Mg^{2+}
3) Synthesis of reducing equivalent.
4) None of the above

17. Read the following statements and choose the incorrect option.
1) Aerobic respiration requires O_2 supply.
2) Krebs' cycle is also referred to as aerobic respiration.
3) The conversion of PGA to BPGA is an energy yielding process and this energy is trapped by the formation of AATP.
4) All of the above

18. Which is true for glycolysis?
1) ATP molecules are synthesized between fructose-6- phosphate and fructose-1,6-biphosphate.
2) Water molecules are released between PEP and pyruvate.
3) NAD^+ is oxidized between PGAL and BPGA.
4) ATP molecules are synthesized in the conversion of BPGA into PGA.

19. In glycolysis, ATP is synthesised during conversion of
1) PEP to pyruvic acid
2) BPGA to PGA
3) 1,3-bisphosphoglyceric acid to 3-phosphoglyceric acid
4) All of the above

Fermentation

20. During the process of anaerobic respiration,
1) A large quantity of CO_2 is always evolved
2) There is net gain is 38 ATP molecules from one glucose molecule
3) Complete oxidation of glucose
4) At least one of the end products is organic

21. Select incorrect statement for the process of fermentation by lactic acid bacteria:
1) Only less than 7% of the energy in glucose is released.
2) The process is hazardous to bacteria due to its end product.
3) Pyruvate is changed to ethanol and CO_2.
4) Only two molecules of ATP are gained per molecule of glucose.

22. Fermentation takes place under anaerobic conditions in
1) Many prokaryotes
2) Unicellular eukaryotes
3) Germinating seeds
4) All of the above

23. Alcoholic fermentation differs from lactic acid fermentation in
1) Oxidation of NADH to NAD^+
2) Net gain of 2ATP molecules
3) Involving decarboxylase activity
4) More than one option is correct

24. Which is true for fermentation?
1) Partial breakdown of glucose
2) Net gain of only 2 ATP
3) Occurs in the cytoplasm of the cell
4) All of the above

25. Read the given assertion and reason carefully and select the correct option.
Assertion: In both lactic acid and alcohol fermentation not, much energy is released.
Reason: Less than seven per cent of the energy in glucose is released and not all of it is trapped as high energy bonds of ATP.
1) If both assertion and reason are true and reason is the correct explanation of the assertion
2) If both assertion and reason are true but reason is not the correct explanation of the assertion

3) If assertion is true but reason is false
4) If both assertion and reason are false

26. Which of the following enzyme is not required during fermentation in microbes?
1) Alcohol dehydrogenase
2) Pyruvate dehydrogenase
3) Lactic acid dehydrogenase
4) Pyruvate decarboxylase

27. The metabolic fate of pyruvate depends upon
1) Cellular need
2) Availability of oxygen
3) Organism
4) All of these

28. Which of the following comparative account is incorrect w.r.t. respiratory process?

	Fermentation	**Aerobic respiration**
1)	Partial oxidation	Complete oxidation
2)	Low amount of energy is released	High amount of energy is released
3)	No loss as heat	Heat loss occurs
4)	No ETC	ETC present

29. During vigorous exercise, when oxygen is inadequate for cellular respiration, reduction of pyruvic acid is catalyzed by
1) Pyruvate dehydrogenase
2) Lactate decarboxylase
3) Lactate dehydrogenase
4) Pyruvate decarboxylase

Aerobic Respiration

30. The TCA cycle starts with the condensation of
1) Acetyl group with oxaloacetic acid (OAA)
2) Acetyl group with phosphoenol pyruvate (PEP)
3) Acetyl group with citric acid
4) Oxaloacetic acid (OAA) with citric acid

31. Which of the following statements is not associated with the breakdown of respiratory substrate under aerobic conditions?
1) The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2.
2) Pyruvic acid undergoes oxidative decarboxylation.
3) Reduction of pyruvic acid in muscles cells.
4) Occurs inside the mitochondria.

32. Substrate entrant of TCA cycle is
1) 3-carbon compound formed in cytoplasm

2) Formed in mitochondrial matrix through oxidative decarboxylation of pyruvic acid
3) Two carbon compound acetyl-CoA
4) Both (2) and (3)

33. The gateway reaction of aerobic respiration
(i) Is also called link reaction
(ii) Is catalyzed by pyruvate dehydrogenase complex
(iii) Involves carboxylation and dehydrogenation
(iv) Produces acetyl-CoA and reducing power
Correct statements is/are
1) Only (i)
2) Only (i) and (ii)
3) (i), (ii) and (iv)
4) (i), (ii), (iii) and (iv)

34. Conversion of citrate into isocitrate in Krebs cycle occurs with the help of enzyme ____ and cofactor ____, respectively
1) Aconitase
2) Fumarase
3) Isocitrate dehydrogenase
4) Citrate synthase

35. The continued oxidation of acetyl-CoA via Kreb's cycle requires the replenishment of
1) α-Ketoglutaric acid
2) Succinie acid
3) Malic acid
4) Oxaloacetic acid

36. The five-carbon intermediate compound formed between the conversion of isocitrate to succinyl CoA.fs
1) α-Ketoglutaric acid
2) Malic acid
3) Succinic acid
4) Oxalosuccinate

37. In TCA cycle, GTP is formed during conversion of
1) α-Ketoglutaric acid into succinyl-CoA
2) Succinyl-CoA into succinic acid
3) Succinic acid into OAA
4) Citrate into isocitrate

38. Pyruvate dehydrogenase catalyzes the conversion of
1) Glucose $\rightarrow$Pyruvate
2) Pyruvate $\rightarrow$ Acetyl-CoA
3) Pyruvate $\rightarrow$ PEP
4) PEP $\rightarrow$ Pyruvate

39. The complex of ETS which has two copper centers associated with it is

1) Succinate dehydrogenase
2) Cytochrome c oxidase complex
3) Cytochrome bc_1 complex
4) Cytochrome reductase complex

40. Terminal electron acceptor of electron transport is

1) CO_2
2) O_2
3) Pyruvic acid
4) Glucose

41. In mitochondrial ETC, cytochrome c is a A protein attached to the B surface of inner membrane.

	A	**B**
1)	Small	Outer
2)	Large	Outer
3)	Small	Inner
4)	Large	Inner

42. Respiratory balanced sheet prepared for oxidation of one molecule of glucose is based on certain assumptions. Read the following statements and state true (T) or false (F).

(A) Sequential orderly functioning pathway including glycolysis, TCA cycle, and ETS.

(B) Only glucose is being respired, no other alternative substrate is entering the pathway.

(C) NADH synthesised in glycolysis is transferred into mitochondria and undergoes oxidative phosphorylation.

(D) The intermediates of TCA cycle are used in different anabolic pathways.

	A	B	C	D
1)	T	T	T	F
2)	T	T	F	F
3)	T	F	T	F
4)	T	F	F	F

43. Which intermediate product of Krebs' cycle is withdraw for synthesis of chlorophyll?.

1) Succinyl-CoA
2) Acetyl-CoA
3) α-Ketoglutaric acid
4) Oxalosuccinate

44. Intermediate common to fatty acid and carbohydrate oxidation is _____.

1) Pyruvate
2) Acetyl-CoA
3) Succinate
4) Oxaloacefate

45. Complete oxidation of one molecule of glucose yields

1) 30 ATP
2) 38 ATP
3) 24 ATP
4) 34 ATP

46. The protein enters respiratory pathway as

1) Pyruvate
2) Glycerol
3) α-Ketoglutaric acid
4) Dihydroxyacetone phosphate

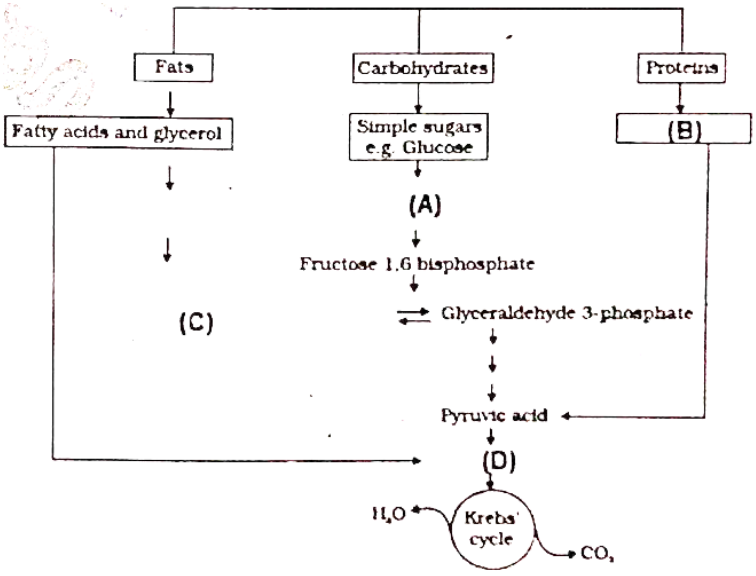
47. It is possible to make calculations of the net gain of ATP for every glucose molecule oxidised; but in reality this can remain only a theoretical exercise.

These calculations can be made only on certain assumptions that:

i. There is a sequential, orderly pathway functioning, with one substrate forming the next and with glycolysis, TCA cycle and ETS pathway following one after another.
ii. The NADH synthesised in glycolysis is transferred into the mitochondria and undergoes oxidative phosphorylation
iii. None of the intermediates in the pathway are utilised to synthesise any other compound.
iv. None of the intermediates in the pathway are utilised to synthesise any other compound.

1) i and iv
2) ii and iii
3) i, ii and iii
4) All the above

48. Choose the correct option w.r.t. given figure.



	A	B	C	D
1)	Glucose-6-phosphate	Amino acid	DHAP	AcetylCoA

2)	Glucose	Amino acid	DHAP	AcetylCoA
3)	Glucose-6-phosphate	Glyceraldehyde 3-phosphate	AcetylCoA	PEP
4)	Glucose	Glyceraldehyde 3- phosphate	DHAP	AcetylCoA

49. In some succulent plants like Opuntia, the RQ is zero because
1) CO_2 is released without any absorption of O_2
2) Oxygen is absorbed but CO_2 is not released
3) Rate of photosynthesis is equal to rate of respiration
4) More than one option is correct

50. RQ (respiratory quotient)
A. Is the ratio of the volume of CO_2 evolved to the volume of oxygen consumed in respiration.
B. Depends upon the type of respiratory substrate used during respiration.
1) Only (A) is correct
2) Only (B) is correct
3) (A) and (B) both are correct
4) (A) and (B) both are incorrect

51. RQ of germinating castor seed is
1) Unity
2) Less than unity
3) More than unity
4) Infinite

52. Respiratory quotient (RQ) or respiratory ratio is the
1) Ratio of the volume of O_2 evolved to the volume of CO_2 consumed in respiration
2) Ratio of the volume of CO_2 consumed to the volume of O_2 evolved in respiration
3) Ratio of the volume of CO_2 evolved to the volume of O_2 consumed in respiration
4) Ratio of the volume of O_2 consumed to the volume of O_2 evolved in respiration

53. $2C_{51}H_{98}O_6 + 145O_2 \rightarrow 102\,CO_2 + 98\,H_2O +$ Energy The RQ of the above reaction is
1) 1
2) 0.7
3) 1.45
4) 1.62

54. RQ of glucose is
1) 1
2) 0.5
3) 2
4) Infinite

55. The respiratory quotient depends upon the
1) Respiratory substrate used during respiration
2) Concentration of the substrate
3) Nature of bonds present in between the molecules of each substrate
4) All of the above

56. The volume of oxygen used is more than the volume of CO_2 evolved for which of the following respiratory substrate?
1) Glucose
2) Protein
3) Tripalmitin
4) More than one option is correct

57. Which of the following respiratory substrate has RQ value more than one?
1) Tripalmitin
2) Malic acid
3) Oxalic acid
4) Both (2) and (3)

58. Read the given assertion and reason carefully and select the correct option.
Assertion: When carbohydrates are used as substrate and are completely oxidised, the RQ will be 1.
Reason: Equal amounts of CO_2 and O_2 are evolved and consumed, respectively.
1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

Answer Keys

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
3	3	2	4	4	1	1	1	3	1	3	2	4	2	4
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
2	3	4	4	4	3	4	3	4	1	2	4	3	3	1
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
3	4	3	1	4	1	2	2	2	2	1	1	1	2	2
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**		
1	4	1	2	3	2	3	2	1	1	4	4	1		